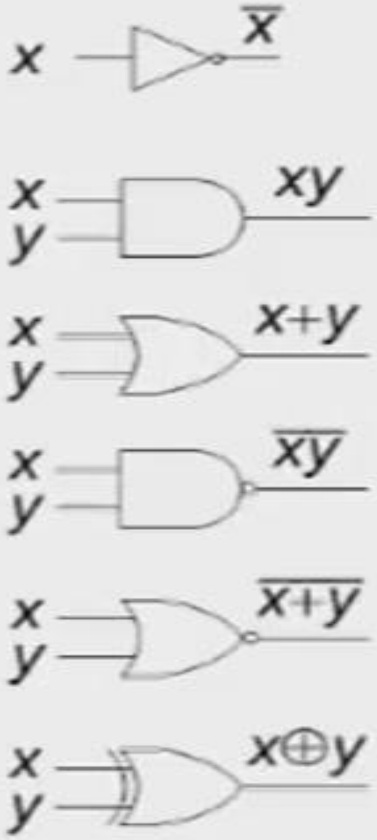
Basic Logic Gates

- Not
- And
- Or
- Nand
- Nor
- Xor



where $\overline{x} = \sim x \quad \neg$

where $xy = x \wedge y$

where $x+y = x \vee y$

where $\sim(xy) = \overline{xy}$

$(p \vee q) \wedge (\neg q)$

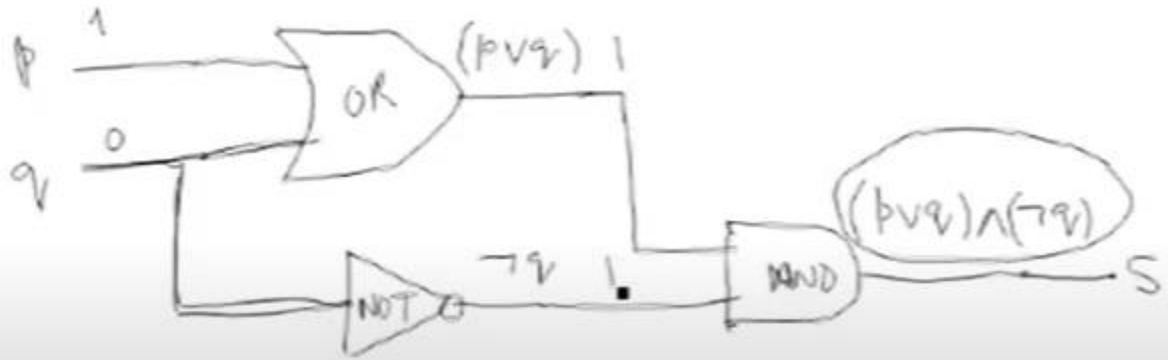
p, q

$\neg$, ( )

$\wedge, \vee$

$\rightarrow$

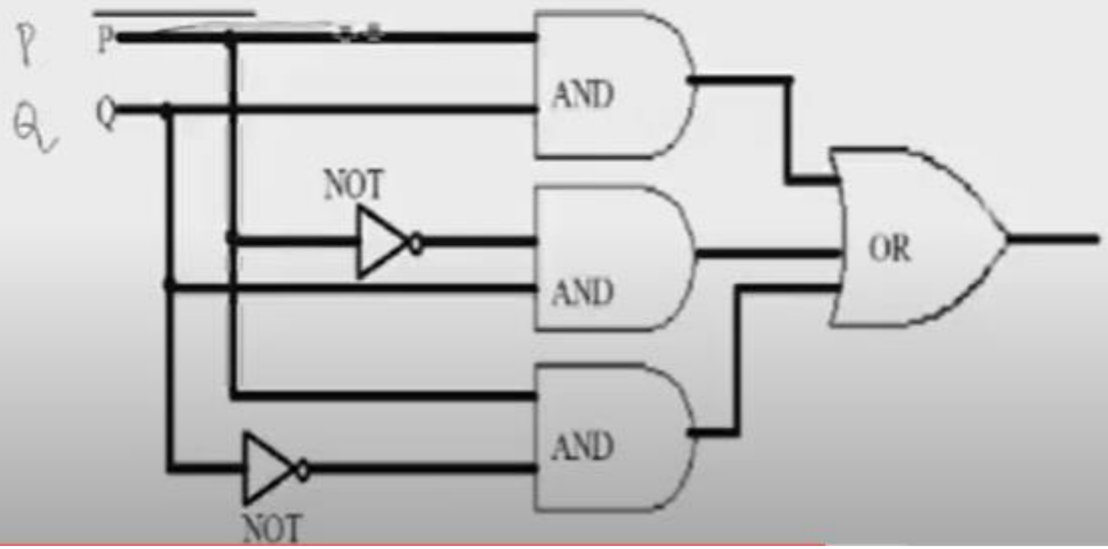
$\leftrightarrow$



Constructing Circuits

Here is the circuit of the statement

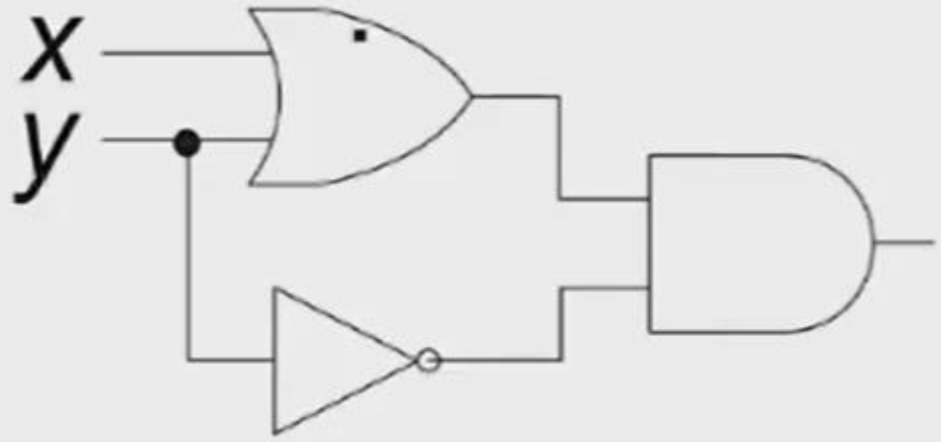
$$(p \wedge q) \vee (\sim p \wedge q) \vee (p \wedge \sim q)$$



Cont...

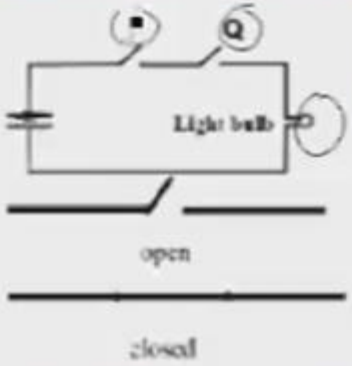
Following is the circuit output of the following statement

$(x + y) \wedge \sim y \equiv \underline{(x \vee y)} \wedge \underline{(\neg y)}$



Switches in series

- We can see the application of logic in switches:



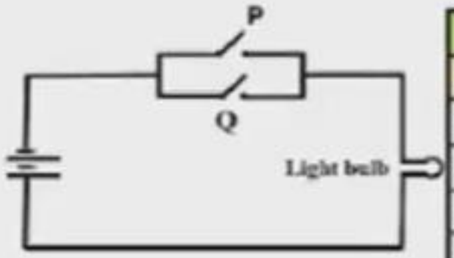
Switches		Light Bulb
P	Q	State
Closed	Closed	On
Closed	Open	Off
Open	Closed	Off
Open	Open	Off

- Here is the truth table of the switch

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Switches in parallel

- We can see the application of logic in switches:



Switches		Light Bulb
P	Q	State
Closed	Closed	On
Closed	Open	On
Open	Closed	On
Open	Open	Off

- Here is the truth table of the switch:

P	Q	P ∨ Q
T	T	T
T	F	T
F	T	T
F	F	F

Designing a circuitt for a given input/output

Here is the out put

INPUTS			OUTPUT
P	Q	R	S
1	1	1	0
1	1	0	1
1	0	1	0
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

we can write it as following

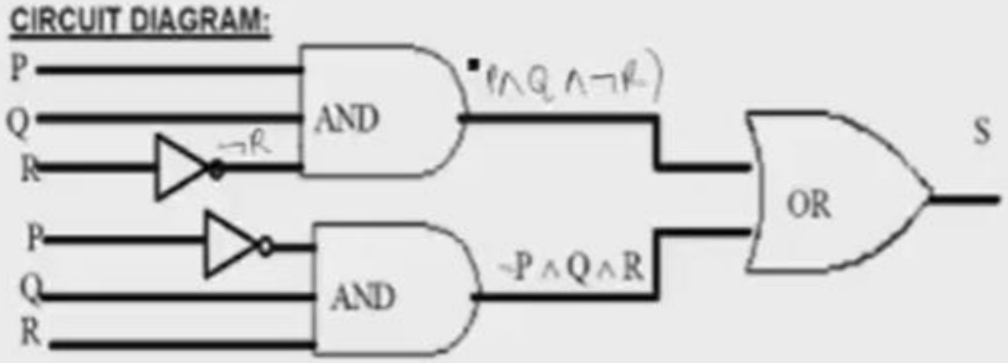
INPUTS			OUTPUT	
P	Q	R	S	
1	1	1	0	
1	1	0	1	$\rightarrow P \wedge Q \wedge \neg R$
1	0	1	0	
1	0	0	0	
0	1	1	1	$\rightarrow \neg P \wedge Q \wedge R$
0	1	0	0	
0	0	1	0	
0	0	0	0	

Designing a circuitt for a given input/output

$(P \wedge Q \wedge (\neg R)) \vee (\neg P \wedge Q \wedge R)$

- Here is the circuit of the previous input/output

P, Q, R.



Circuit reductions

Following is the circuit representations of the statement

$$(P \wedge Q) \vee (\sim P \wedge Q) \vee (P \wedge \sim Q)$$